

Design and Evaluation of a Transimpedance-amplifier for a LiFi Receiver

Research proposal for Master's Project

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1 Introduction

Li-Fi(Light Fidelity) is a new technology and an alternative method for wireless communication, also known as Visible Light Communication(VLC). Generally, an AFE (Analog Front-End) circuit of a VLC system is divided into two parts: i) transmitter and ii) receiver. The transmitter consists of an analog/digital driver circuit, and the receiver includes a photo-detector (photodiode or phototransistor), TIA and specific analog circuit (amplifiers, filters and other circuit configurations)[5].

TIA(Transimpedance-Amplifier) are generally used to amplify the output of current-based sensors like photodiodes, also called photodiode amplifiers. It has a function for converting the photodiode photocurrent output to a voltage signal. TIA has been implemented in various optic applications for optical sensing (measurement purposes) and optical communication (i.e. IR, VLC, fiber-optic) [5].

1.1 Motivation

Although TIA has been in use since the 1970s, most research describes its detailed design and implementation for precision measurement. More research is still needed to cover the TIA design for LiFi applications. When dealing with small components and variable distances, reducing the overdrive of the operational amplifier (OPA) is crucial for maintaining speed, especially with high data rates and a significantly high amplifier gain.

1.2 Scope of work

If low power and high speed are targeted, higher noise is obtained at TIA; therefore, we need special requirements are needed. In this project, research, analysis, and comparison of different components and circuits for a TIA will be done. These can be met by selecting specific OP-AMPs and advanced electronic configuration circuitry. Discussions in this paper are limited to the focus on photodiode and trans-impedance Amplifiers only.

Firstly, TIA's general topology will be evaluated, and then all of the usage components will be verified using three approaches: mathematical calculation, simulation, and actual experiment. The primary object would be to tackle the problem of OP-Amp overdrive with dynamic distances. For simplicity of the expose and understanding of further work, some design requirements have been gathered below:

- Supply Voltage: 5V
- Supply Current: <2mA
- Input Current: 0-90 μ A
- Output: 100mV-4.9V

2 Related Work

Designing a single-supply photodiode amplifier requires considering many factors, including stability and input and output voltage range limitations. Furthermore, the effects of DC error sources, such as input bias current and input offset voltage, are often ignored and can degrade the accuracy of the circuit [4].

The related work can be divided into four sections. First, the background of the VLC platform is as the primary reference, followed by Photodiode Mode Operations and its Equivalent circuit. Thirdly, the circuit design of TIA operation is crucial because the success of the rest of the project depends on it, and any error at this stage will affect the performance of the TIA. A considerable amount of research is needed in this part, which consists of several subsections: Gain and Capacitor Feedback Calculations, Amplifier Gain Bandwidth Calculation, and Bias Network calculation. Lastly, a considerable amount of research needs to be done to do noise analysis and compare different components.

2.1 Lifi System

An extensive description of a Li-Fi system has been presented in the paper [5]. TIA has been implemented in various optical applications for optical sensing (measurement purposes) and optical communication (i.e., IR, VLC, fiber-optic). It can be built by discrete circuit mechanisms or MOS transistor circuitry. Its circuit design is divided into several aspects, such as precision instrumentation TIA, high-speed TIA, low noise TIA [8], gain stability TIA [3], manual programmable TIA [2], and other related objectives.

This paper[5] considers a point-to-point VLC design with one link with $> 1\text{MHz}$ bandwidth(target estimated). Since OFDM(Orthogonal Frequency Division Multiplexing) is a multicarrier signal, the system is assumed to transmit over 10Mb/s. In order to achieve this target, the received signal in the receiver (Rx) must be processed by low error (correctly), and an available commercially analog component is chosen providently. A high-brightness white LED with a high Field of View (FOV) is chosen for wide light distribution. Photodiode was selected as a photo-detector because it has a fall-rise time better than photovoltaic, photo-transistor, and Light Dependent Resistor (LDR).[5]

2.2 Photodiode Mode Operations

The paper [5] considers the photoconductive mode because it has low noise and precision characteristics. Therefore, an information signal emitted from an LED can be received with a minimum noise level without adding specific circuitry to minimize noise. That also means the photodiode is in reverse bias, and applying a reverse bias voltage to a pn junction causes the depletion region to become wider. Thus, the photoconductive mode allows for broader bandwidth and is preferable when producing more output signals relative to illuminance. In addition, a wider depletion region reduces the photodiode's junction capacitance, which maximizes the detector's ability to respond to rapid variations in illuminance, which is suitable for Photodetection/Optical communication [6]. But this

Table 1: Selected photodiode parameters

Parameter	Condition	Symbol	Min	Typ	Max	Unit
Diode Capacitance	$V_R = 0V$	C_J				pF
Reverse Light Current	$E_e = 1mW/cm^2$	I_{PD}				μA

mode is more prone to noise/Dark current.[6] So, filtration is necessary.

2.3 Photodiode Parameters

In the paper [1], OSRAM SFH213 has been used, and in this design, the photodiode is minimally reverse biased, and therefore, the junction capacitance when $V_R = 0V$ will be used for stability calculations. Also, the effects of dark current on the amplifier output can be neglected because of the minimal reverse bias. Parameters of the photodiode are shown in Table 1. These parameters are important for the selection of an optimum photodiode, according to the author [1]. The researcher of the paper[4] targeted UV to NIR precision photometry applications. For the circuit, they used Hamamatsu’s S1336-5BK Si photodiode. According to them, Shunt Resistance is also an important parameter to consider. The circuit in paper [5] is based on the SP8-ML KODENSHI photodiode and took real experiment value of short current/Reverse Light Current.

Three different types of photodiodes are used in the paper [4] for noise analysis. All of those are manufactured by HAMAMATSU, whose types are Si PIN (320 nm–1100 nm), Si APD (200 nm to 1000 nm), and Si (380 nm to 780 nm). They choose three different types of photodiode, especially with significant differences in pF parameter, i.e., 3 pF, 30 pF, and 150 pF. The purpose is to find the effect of capacitance on noise.

2.4 Circuit Description

Properly designing a single-supply photodiode amplifier requires considering many factors, including stability and input and output voltage range limitations. Furthermore, the effects of DC error sources, such as input bias current and input offset voltage, are often ignored and can degrade the transient response of these circuits[1].

This design in paper[1] examined the proper design process for photodiode amplifiers used in single-supply applications. The photodiode is operated in photoconductive mode: exposure to light causes a reverse current through the photodiode. The diode is connected such that this current causes the op-amp output voltage to increase. A feedback capacitor is necessary for most photodiode amplifiers to maintain stability. This capacitor compensates for the photodiode capacitance at the inverting input of the op-amp[9]. A small bias voltage is derived from the positive supply and applied to the op amp’s non-inverting input. This prevents the output from saturating at the negative supply rail without input current[9].

According to both papers [1] and [9], without any photodiode current, the amplifier output will attempt to settle at the voltage applied to the non-inverting input. If the non-inverting input is grounded, the output voltage would ideally be 0V. However, the

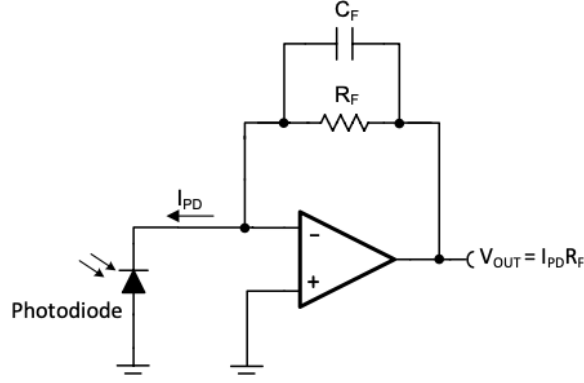


Figure 1: Connection of a photodiode to the transimpedance amplifier (photoconductive mode).

output voltage can never reach 0V because this is also the value of the negative power supply. Therefore, the op-amp output will “saturate” near its negative power supply. This undesirable condition can delay the amplifier’s response to an input signal. In order to avoid saturation, a resistor divider from the positive supply is used to bias the amplifier input above ground in Fig. 2.

Then, the output transfer function, including the bias voltage, becomes:

$$V_{OUT} = i_{PD}R_1 + V_B = i_{PD}R_1 + V_{CC}\frac{R_3}{R_3 + R_2} \quad (1)$$

2.4.1 Bias Network

The photodiode acts as a reverse-biased diode at DC, so the gain from the non-inverting terminal of the op-amp is 1. Therefore, when the photodiode current is 0A, $V_{OUT} = V_{Bias}$ [9]. Paper [9] considers that the bias voltage of 200mV is derived from the positive supply and applied to the non-inverting input to prevent the amplifier output from saturating at the negative power supply rail. The equation gives the bias at the non-inverting input, according to Fig.3 :

$$V_{Bias} = V_{CC}\frac{R_{B2}}{R_{B1} + R_{B2}} \quad (2)$$

Then R_{B1} :

$$R_{B1} = \frac{R_{B2} * (V_{CC} - V_{Bias})}{V_{Bias}} \quad (3)$$

If the photodiode current is 0 A, the amplifier with a noise gain of 1 attempts to produce a 0V output, resulting in the amplifier saturating and its bandwidth being compromised due to the saturation recovery time. It is best to account for the V_{OUT} requirement of the amplifier by adding a bias voltage that defaults the TIA output above the maximum V_{OUT} from the negative rail[9].

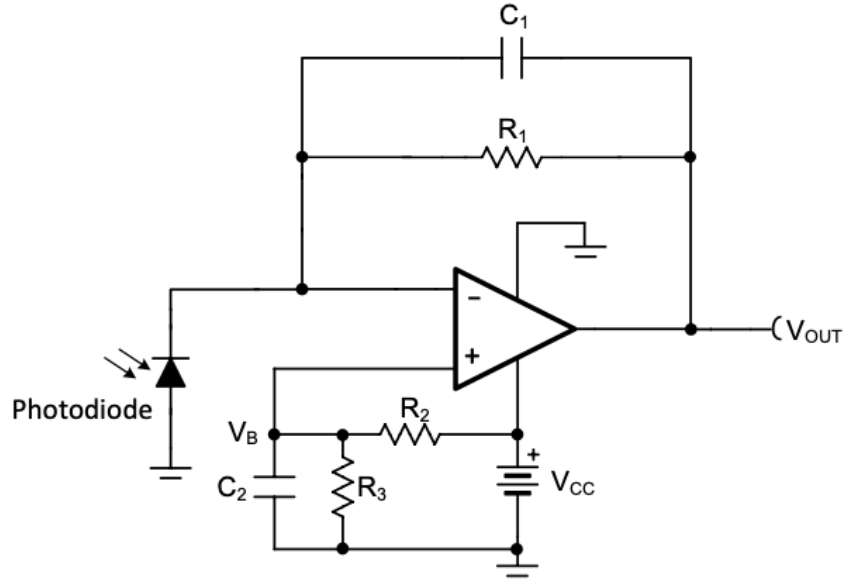


Figure 2: A bias voltage is applied to the op amp's non-inverting input.

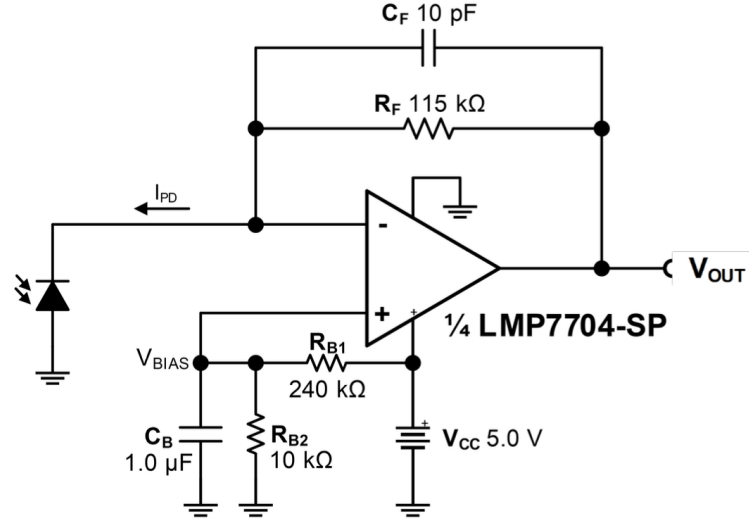


Figure 3: SpaceGrade TIA with Bias Circuit

It is best to account for the V_{OUT} requirement of the amplifier by adding a bias voltage that defaults the TIA output above the maximum V_{OUT} from the negative rail. For many rail-to-rail input/output (RRIO) op amps, the typical output swing limitations range from 30 to 60 mV depending on operating conditions and in some instances, can be as high as 200 mV. A bias voltage V_{Bias} of 200 mV accounts for the worst-case scenario[9].

Paper [1] considers that the bias voltage of 100mV. The equation gives the bias at the non-inverting input, according to Fig.2 :

$$V_B = V_{CC} \frac{R_3}{R_3 + R_2} \quad (4)$$

For a 100mV bias voltage:

$$.1 = 5 \frac{R_3}{R_3 + R_2} \quad (5)$$

$$R_2 = 49R_3 \quad (6)$$

The closest 1 percent resistor values conform to this ratio is $R_2 : 13.7K\Omega$, $R_3 : 280\Omega$.

Capacitor C_2 is placed in parallel with resistor R_3 to reduce the noise contribution of the resistor divider and prevent power supply noise from affecting the amplifier output. Selecting a value of $1\mu F$ for C_2 produces a corner frequency of:

$$f_p = \frac{1}{2\pi C_2(R_2 || R_3)} = \frac{1}{2\pi * 1\mu F * (13.7K\Omega || 280\Omega)} = 580.028Hz \quad (7)$$

This corner frequency should be low enough to prevent power supply noise from passing to the output [1].

2.4.2 Gain and Capacitor Feedback Calculations

To determine the maximum trans-impedance gain, papers [4],[1] and [9] consider two factors: the maximum reverse current, $I_{IN(MAX)}$ and the desired output voltage range, $V_{OUT(MAX)}$ and $V_{OUT(MIN)}$.

The value of the feedback resistor, which sets the trans-impedance amplifier (TIA) gain [4] [9], can be calculated by dividing the maximum output voltage(limited by the amplifier's power supply) by the maximum input current:

$$\frac{V_{OUT(MAX)} - V_{OUT(MIN)}}{I_{IN(MAX)}} = R_1 \quad (8)$$

The circuit calculation in paper [5] took the real experiment value of short current/Reverse Light Current instead of the maximum value based on the SP8-ML KODENSHI photodiode datasheet.

The feedback capacitor, in combination with the feedback resistor, forms a pole in the frequency response of the amplifier [9]:

$$f_p = \frac{1}{2\pi C_1 R_1} \quad (9)$$

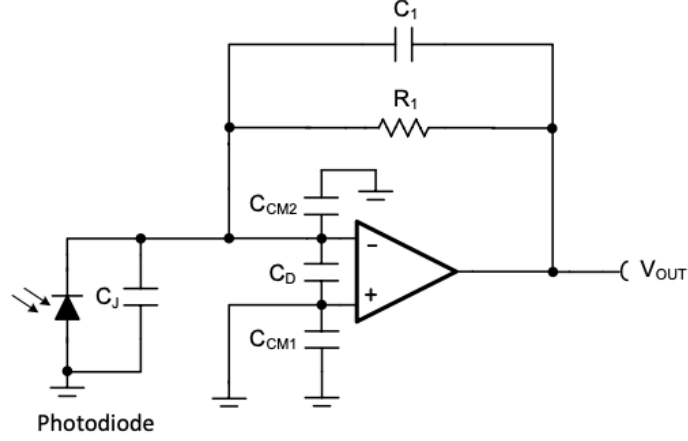


Figure 4: Transimpedance amplifier circuit showing capacitances at the inverting node.

Above this pole frequency, the amplification of the circuit will decline. The maximum feedback capacitor value can be determined from the feedback resistor value and the desired bandwidth:

$$C_1 \leq \frac{1}{2\pi f_p R_1} \quad (10)$$

Keeping the feedback capacitor at or below the value calculated in Equation 10 ensures that the circuit meets the bandwidth requirements[1]. On the other hand, to ensure the circuit meets the 1 MHz bandwidth, in papers [5] and [4], equation 11 is used. Capacitors play a crucial role in minimizing noise, optimizing TIA response, ensuring TIA circuit stability, and reducing overshoot. The larger the C_1 , the slower the response of the TIA and the smaller the C_1 , so that the amplifier oscillates, even though it increases the bandwidth[4]. The value of C_1 forms a pole in the frequency response with 45° of phase margin, which is given by:

$$C_1 = \sqrt{\frac{C_J + C_D}{2\pi f_{GBP} R_1}} \quad (11)$$

2.4.3 Amplifier Gain Bandwidth Calculation

After calculating the maximum feedback capacitor value allowable to meet the design requirement for bandwidth, it is necessary to calculate the necessary op amp gain bandwidth for the circuit to be stable[1]. In Figure 4, the circuit from Figure 2 has been redrawn to show the junction capacitance of the photodiode(C_J) and the differential (C_D) and common-mode (C_{CM1}, C_{CM2}) input capacitances of the amplifier. The bias voltage applied to the non-inverting input is considered an AC ground[9]. From this illustration, it is apparent that C_J, C_D , and C_{CM2} are in parallel, and the capacitance at the inverting input is:

$$C_{IN} = C_J + C_D + C_{CM2} \quad (12)$$

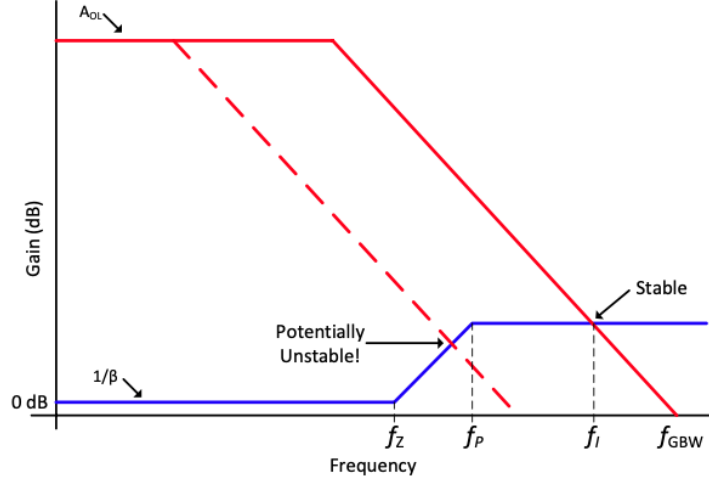


Figure 5: The open loop gain (red) and inverse feedback factor (blue) curves of a typical transimpedance amplifier.

The AC ground shorts C_{CM1} at the non-inverting input and does not contribute to the input capacitance calculation.

2.4.4 Stable Amplifier Feedback

A rate-of-closure (ROC) analysis is one method to infer the stability of a system graphically by observing the closure rate between the AOL and $(1/\beta)$ where they intersect. These curves should intersect with a 20dB/decade rate-of-closure for a stable system[1]. This paper also provided an example. Figure 5 shows a typical trans-impedance amplifier's open loop gain (AOL) and inverse feedback factor $(1/\beta)$ curves. suppose the AOL curve of an amplifier decreases at 20dB/decade, and the $(1/\beta)$ curve increases at 20dB per decade at the point of intersection. In that case, the closure rate is 40dB/decade, and the system is potentially unstable. The capacitance at the inverting input of the op-amp produces a zero in the $(1/\beta)$ curve at:

$$f_z = \frac{1}{2\pi(C_1 + C_{IN})R_1} \quad (13)$$

Above this zero, the $(1/\beta)$ curve will rise at 20dB/decade. If the AOL curve intersects the $(1/\beta)$ curve in this rising portion (as illustrated by the dashed line in Figure 5), the system will potentially be unstable[1]. For the amplifier to be stable, the AOL curve should intersect the $(1/\beta)$ curve after the pole created by the feedback capacitor:

$$f_p = \frac{1}{2\pi C_1 R_1} \quad (14)$$

In the region above the pole, the frequency of intersection is:

$$f_p = \frac{C_1}{C_{IN} + C_1} f_{GBW} \quad (15)$$

Where (f_{GBW}) is the unity gain bandwidth of the op-amp. Therefore, in order to ensure stability, the unity bandwidth of the op-amp must obey the rules:

$$f_1 > f_P \quad (16)$$

Inserting the equations for (f_1) and (f_p) into this rule and solving for unity gain bandwidth, we arrive at a useful equation:

$$\frac{C_1}{C_{IN} + C_1} f_{GBW} > \frac{1}{2\pi C_1 R_1} \quad (17)$$

$$f_{GBW} > \frac{C_1 + C_{IN}}{2\pi C_1^2 R_1} \quad (18)$$

C_{IN} must first be determined to calculate this design's unity gain bandwidth requirement. C_D and C_{CM2} will be known from the data sheet of selected specific op-amp[1].

2.4.5 Noise Analysis

According to paper [4], to determine the peak noise, we can use Equation (19). This assumes that the first zero and pole of the noise density output are less than a decade lower than the second pole and that the output noise is equal to the plateau noise ($N_2 = \text{plateau noise}$). Here, " e_n " refers to the voltage noise density obtained from the datasheet.

$$N_2 = e_n \left(\frac{C_j + C_{in} + C_1}{C_1} \right) \quad (19)$$

The calculation of N_V is based on Equation (20), where (N_2 is obtained from Equation (19) and (f_{p2} are obtained from Equation (21). Then the N_V can be rewritten as Equation (22), The total sum of C_j , C_{in} , and C_1 is C_{tot} or C_t in Farad.

$$N_V = N_2 \sqrt{\left(\frac{\pi}{2} * f_{p2} \right)} \quad (20)$$

$$f_{p2} = f_{GBW} \left(\frac{C_1}{C_j + C_{in} + C_1} \right) \quad (21)$$

$$N_V = e_n \left(\frac{C_j + C_{in} + C_1}{C_1} \right) * \sqrt{\frac{\pi}{2} * f_{GBW} \left(\frac{C_1}{C_j + C_{in} + C_1} \right)} \quad (22)$$

The current noise of the Op Amp will be present on the output after passing through R_1 , which can be calculated using Equation (23) [4]. The input noise density I_n for each Op Amp can be obtained from its datasheet. Equation (24) is used to determine the equivalent noise bandwidth (ENBW). Equation (25) can be used to calculate the cut-off frequency (f_{-3dB}) of TIA.

$$N_{current} = I_n * R_1 * \sqrt{ENBW} \quad (23)$$

Table 2: Basic op amp requirements

Parameters	Values
Minimum Supply Voltage	$< 5V$
Quiescent Current	$< 2mA$
Input Voltage Range	$(V-)+0.1$
Output Voltage Range	$(V-)+0.1, (V+)-0.1$
Gain Bandwidth Product(GBWP)	$> 9.653 \text{ MHz}$

$$ENBW = f_{-3db} * \frac{\pi}{2} \quad (24)$$

$$f_{-3dB} = \sqrt{\frac{f_{GBW}}{2\pi * R_1(C_j + C_{in})}} \quad (25)$$

The noise in R_1 of TIA is given by Equation (26). Where k is the Boltzmann (1.38×10^{-23}) constant and T is the temperature in Kelvin (283 K).

$$N_{resistor-feedback} = \sqrt{4kT * ENBW * R_1} \quad (26)$$

There are three independent sources of noise in the mathematical calculation[4]. If N_{total} is Gaussian, it means that it is the root-sum-square (RSS) given by Equation (27). If f_p is much higher than the signal bandwidth, N_{total} can be significantly reduced by adding single-pole RC filters to the specific circuit of the Low-Pass Filter (LPF) on the TIA output[4].

$$N_{Total} = \sqrt{N_{resistor-feedback}^2 + N_{current}^2 + N_{voltage}^2} \quad (27)$$

2.5 Component Selection

2.5.1 Amplifier

The author of the paper[1] details the basic requirements for the op-amp used in its application in Table 2.

In this experiment of paper [1], the reason for selecting the OPA320 for this design was due to its low-bias current, offset voltage, power consumption, and wide bandwidth. The design of OPA320 satisfies all the necessary specifications. The combined capacitance values of the differential and common mode inputs are estimated to be around 9pF, slightly less than the expected value. However, the gain bandwidth product is still sufficient for a stable design. The slew rate of the OPA320 limits the circuit's full power bandwidth, which is the highest frequency sinusoid produced at the amplifier's full output swing [1].

Texas Instrument (TI) Inc. manufactures all the operational amplifiers used in the paper [5]. There are three main reasons for using products from TI Inc.: the SPICE® circuit files that are available for most of their products, the sampling option they offer,

and the availability of specific simulation tools to prove the design; in this term, it uses *TINATM* software. Rise and fall time, slew rate, input capacitance, voltage, and current noise density are important parameters when choosing an opamp. OPA656 is chosen, whose unity-gain bandwidth of 500 MHz at gain = 1, low input bias current, low offset and drift, low distortion, high output current, and low input voltage noise make it a great fit for high-speed photodiode amplifier in VLC applications[5].

The paper's researcher [9] selected The LMP7704-SP for this design because of its excellent combination of low bias current (as an op amp with CMOS inputs), offset voltage, power consumption, radiation performance, and wider supply range. This could allow for re-use in other applications in the system and use without a bias voltage in systems with bipolar voltage rails available. LMP7704-SP has a gain bandwidth (GBW) of 2.5 MHz. This value provides sufficient margin to ensure a stable design. However, it's important to note that the op-amp's slew rate (SR) might limit the full power bandwidth of the circuit. The full power bandwidth is the highest frequency sinusoid that can be reproduced at the full output swing of the amplifier.

A more extensive comparison was made between 8 Opamp for noise analysis in the paper [4]. Those manufacturers are Maxim integrated circuits®, Analog Devices®, Texas Instruments®, Linear Technologies®, Microchips®, ROHM®, Intersil®, and Fairchild or ON semiconductor® where each manufacturer produces Op Amp with suitable features to develop the TIA circuit.

2.5.2 Passive Components

Resistors

1 percent resistors were used throughout the design. If greater precision is required, higher tolerance resistors may be used.[1]

Capacitors

The compensation capacitor, C_1 , should be an NP0/C0G ceramic capacitor to avoid affecting the amplifier's frequency response or output distortion. Capacitor C_2 can be a high-k dielectric type with a suitable voltage rating. The decoupling capacitors used on the PCB were also high-k ceramic capacitors.[1]

3 Fundamentals

3.1 Lifi System

Generally, a transmitter (Tx) sends data from a computer and emits a white LED. A photodiode receives the transmitted data, and its output current is converted into voltage. Phototransistors can also detect light signals, but the Photodiode is more recommended because it has better linearity, dynamic range, and stability[4]. The photodiode generates a small current proportional to the light illumination level. A high gain across the TIA is required to overcome weak photocurrent signals when the receiver (Rx) is kept out from the light source[5]. However, with increasingly large

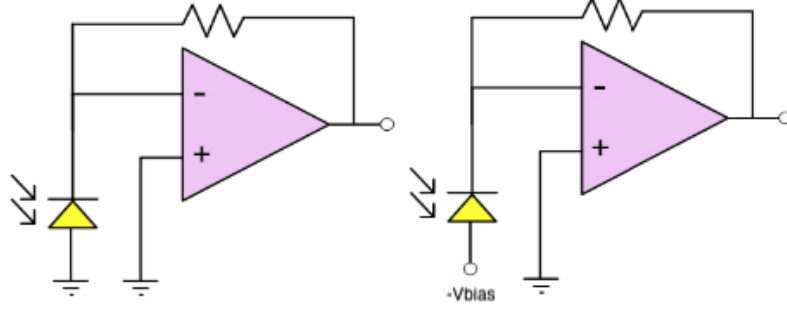


Figure 6: TIA models for photodiode: (a) photovoltaic mode; (b) photoconductive mode

gains, the bandwidth will be narrower. Selecting a photodiode with high current output is typically challenging because it is susceptible to high noise. A possible approach is adding an amplifier circuit. Generally, this Amplifier circuit is connected to analog filters, a circuit for eliminating noise signals from other light interferences [5].

3.2 Photodiode Mode Operations

The basic output of a photodiode is current that flows through the device from cathode to anode and is approximately linearly proportional to illuminance[7]. A series resistor or a current-to-voltage amplifier converts The photocurrent into a voltage for further signal processing[6].

Photodiode has two operation modes, i.e., photoconductive and photovoltaic; each mode has advantages and disadvantages. The photovoltaic mode has several characteristics, such as a) without bias, b) non-generate dark current, c) linear, d) low noise, and e) precision instrumentation compatible. On the other hand, photoconductive modes are: a) reverse bias, b) generate dark current, c) nonlinear characteristic, d) have significant noise, and e) high-speed compatibility. These configuration modes are illustrated in Fig 6[5].

3.3 Amplifier

In its most simple form, a trans-impedance amplifier consists of an op-amp and a feedback resistor, Fig.7. The current to be amplified is applied to the inverting input, causing the output voltage to change according to the equation:

$$V_{OUT} = -I_{IN}R_F \quad (28)$$

3.4 Noise Analysis

Noise modeling and analysis are important in VLC channels [4]. Noise is the most important parameter in determining signal-to-noise ratio (SNR) in the OWC(optical

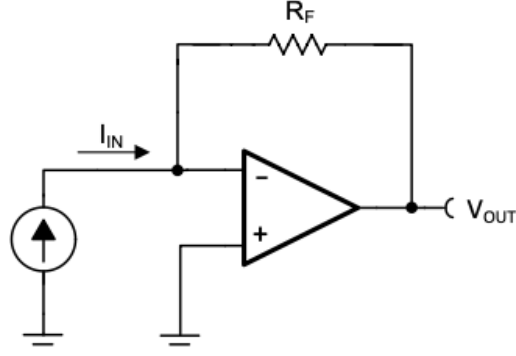


Figure 7: A basic op amp transimpedance amplifier

wireless communication) network. There are two types of noise (i.e., external and internal noise). Internal noise is present in each component of the analog circuits of AFE of the VLC system, which can affect and degrade the quality of signal and data. The Op-Amp, the active component, has natural internal noise known as E_n and I_n , and intrinsic capacitance called C_{IN} . On the other hand, the photodiode, as a sensor device, has an intrinsic capacitance or junction capacitor (C_j) and intrinsic or shunt resistor (R_{sh}). These variables should be considered when selecting components to design the TIA circuit. It's important to note that the larger the values of E_n , I_n , C_{IN} , or C_j , the higher the noise on TIA[4].

4 Evaluation

TIA has an essential and substantial role in optic communication; the function is a converter device, photocurrent to voltage. Emitted data(s) from the LED, as the transmitter device, must be read correctly by a photodiode. Thus, TIA's precision design requires a high bandwidth to support high-speed communications. This work will further calculate the parameters of a TIA circuit for VLC with a general topology design[5]. The input and output voltage range requirements are selected to ensure linear operation over the desired signal amplitudes. According to paper [4], when designing a trans-impedance amplifier (TIA), several criteria related to Op Amps should be considered. These include a) High Bandwidth, b) Low Noise, c) Low power consumption, d) good stability and sensitivity, e) large gain, and f) availability of components in the commercial markets.

In addition to these basic requirements, the op-amp should contribute negligible errors from voltage offsets and noise. Op amps with JFET or CMOS inputs are the most commonly selected type because these op-amps typically have a much lower bias current than those with BJT input devices. This reduces DC error voltages and lowers noise due to reduced input current noise. A complete noise analysis of this amplifier is beyond the scope of this document[1].

It would be useful to investigate the noise behavior in TIA circuits in VLC to predict

the effects of various parameters and characteristics, such as different Op-Amps and photodiodes. This study can be employed in VLC, and trade-offs between complexity, performance, reliability, and cost should be made[4]. There are 14 variables that need to be considered[4]. Those are IC manufacturer, IC series, packaging type, quantity of Op Amp in one chip (channel), intrinsic capacitances of IC (C_{IN}) in pF, input current noise of IC (E_n) in $\text{nV}/\sqrt{\text{Hz}}$, input voltage noise of IC (I_n) in $\text{fA}/\sqrt{\text{Hz}}$, maximum positive supply voltage (V+) to minimum negative supply voltage (V-), output voltage swing capabilities (Vout), gain of IC (AVOL) in dB unit or V/V, gain bandwidth product of IC (GBW) in MHz, slew rate IC in $\text{V}/\mu\text{S}$ unit, power consumption in mWatt and input bias current (I_{Bias}). The cost factor is not considered in this experiment.

To further develop this work, achieving the optimized and crucial requirements is necessary. There are two ways to verify the mathematical calculation of frequency response: one is by simulating the circuit, and another is by doing actual experiments, supposedly with PCB Design[1]. The bode plot, DC transfer response, and Stability Analysis are also needed.

4.1 Simulation

To ensure the correctness of the design, simulation and actual experiments both need to be compared. The datasheet informs us of the typical characteristics of OP-AMP, such as open-loop gain, phase margin, amplifier pulse response, etc.; however, self-experimentation is more convincing in obtaining the best systems. Several simulation tools are in the market, such as LTSpice, PSpice, and TINA-TI™. Simulation in any of them should be adequate. According to the paper[9], these simulations are necessary for properly designing the photodiode amplifier:

- **DC Transfer Function:** This analysis helps determine the transient initial conditions and linearized small-signal models for nonlinear devices. The DC analysis can also produce DC transfer curves. To generate the curves, a specified independent voltage or current source is stepped over a user-specified range, and the DC output variables are stored for each source value sequentially, as well as the circuit's gain and output voltages. In DC transfer function analysis, the input and output are defined in the electronic circuit to calculate the transfer function of small DC signals, and the output/input conversion ratio, input impedance, and output impedance are calculated[**ltspice-dc**]. A DC analysis is generally performed before a transient analysis to determine the transient initial conditions and prior to an AC small-signal analysis to determine the linearized, small-signal models for nonlinear devices[**types**].
- **AC transfer Function:** The AC transfer function was measured using an AC transfer characteristic analysis. AC analysis contains only sinusoids at the same frequency as the input signal. Thus, each signal is represented by only two numbers, one that gives the magnitude and the other that gives the phase. With such a simple signal representation, it is impossible to represent transient behavior [**ac**].

- Stability analysis: For the loop stability simulation, the amplifier's feedback path is broken at the output using a large inductor. This inductor allows the circuit to converge to the proper DC bias point but acts as an open circuit in an AC transfer characteristic simulation. A voltage source is AC coupled into the feedback loop by a capacitor and the loop gain is measured by voltage probe VOUT[1].

Other analyses which are also considerable are:

- Transient analysis: It computes the transient output variables as a function of time over a user-specified time interval. A dc analysis can automatically determine the initial conditions. All sources that are not time-dependent (for example, power supplies) are set to their DC value[types].
- Noise Analysis: The noise analysis is a process that helps analyze the noise generated by a device in a circuit. By considering an input source and an output node, the analysis calculates the noise contributions generated by each device and noise generator within the device to the voltage of the output node. It also measures the input noise level from the particular input source to generate the equivalent output noise. This calculation is performed for every frequency point in a specified range, and the resulting noise value corresponds to the spectral density of the circuit variable viewed as a stationary Gaussian stochastic process. In addition, by combining a DC sweep, noise analysis can compute noise at each point over a range of bias conditions. [types]. In paper[4], calculations of noise is done by equations for different frequencies, photodiodes, feedback resistors, and amplifiers. These plots are done using MATLAB.

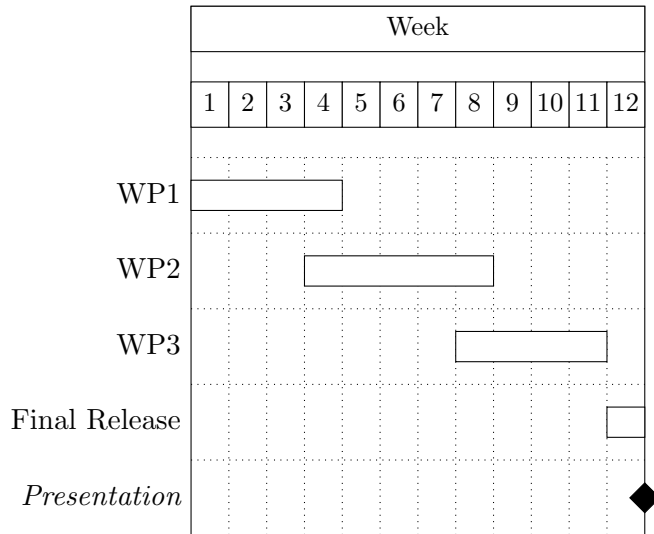
4.2 PCB Design

The TIA's PCB must be careful with lay-outing because the incorrect design can make circuits unstable. After all, parasitic inductance and capacitance generated at high frequency by the PCB trace themselves factor[1] For this reason, the ground pour of the PCB should be removed in the areas around the inverting input of the op-amp on both layers. Matching the output impedance to coaxial cables and measurement equipment prevents ringing in the transient response of wide-bandwidth systems[1]. Of course, verification is necessary for the printed PCB Design. Similar to simulations, these verifications are necessary for properly designing the photodiode amplifier.

In paper [1], the system AC performance was tested by installing the SFH213 photodiode onto the PCB and illuminating it with a light source. A blue LED and 162 ohm resistor were installed onto the output of TIPD108. The output light signal is a square wave at 500kHz.

Under the experimental scheme in paper [5], the TIA board will be fabricated as a double-layer PCB with discrete components. To obtain voltage stability, we use dual power supplies +5 and -5V. An SMA socket is used as a connector from the TIA to an amplifier circuit stage.

5 Schedule



5.1 Work Packages 1

In this work package, the theory of the project needs to be established. Every possible process behind the project principle needs to be considered. The circuit design of our TIA operation is crucial because the success of the rest of the project depends on it, and any error at this stage will affect the performance of the TIA of the final project. A considerable amount of research is needed in this part, which consists of several subsections: Gain and Capacitor Feedback Calculations, Amplifier Gain Bandwidth Calculation, and Bias Network calculation. After the calculation, it needs to be agreed upon that this is the project goal. If agreed upon, optimum circuit components, both active and passive, are selected among different possibilities and providers.

5.2 Work Packages 2

This work package will verify calculated circuit arrangements and components. To do this, circuit configuration will be simulated. The argument will involve which simulation tool will be used. Several simulation tools, such as LTSpice, PSpice, and TINA-TI, are available on the market. DC Transfer Function, AC Transfer Function, Stability Function, Transient Analysis and Noise Analysis are the most important. It may be necessary to go back and forth with the circuit calculation and simulation; some parameters must be changed according to requirements. So both work will be imposed on each other.

5.3 Work Packages 3

This work package is the part where theory is met with practicality. The PCB of the TIA requires careful layout design to ensure circuit stability. This is because

parasitic inductance and capacitance generated by the PCB trace at high frequencies can cause instability. However, designing the PCB can be challenging since it requires learning how to use PCB design tools which have a steep learning curve. Once the PCB has been designed and printed, the entire project should be verified using various electrical instruments. If the PCB circuit board adheres to the calculated and simulated parameters, the project can be deemed a success.

6 Editing

This *research proposal* is the base document for editing the proposed project. It describes the general introduction and motivation of the project. The topic as well as objectives are described and a rough working plan is suggested.

The time span is limited to 3 months.

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